Class 9: Candy Mini Project

Saket Chodavarapu (PID: A18582086)

Table of contents

bg 1
Exploratory Analysis 3
Overall Candy Rankings 7
Adding color 13
Taking a look at pricepercent 15
Exploring the correlation structure 17
Principal Component Analysis 18
Summary 22

bg

ggplot, stats, corr analysis, principal component analysis ##data import csv from 538

```
candy_file <- read.csv(url("https://raw.githubusercontent.com/fivethirtyeight/data/master/ca

candy <- data.frame(candy_file, row.names="competitorname")
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
```

```
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

> Q1. How many different candy types are in this dataset?

There are 85 rows in this dataset.

> Q2. How many fruity candy types are in the dataset?

```r
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types in this dataset.

> Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy in the dataset is Air Heads. Its winpercent value is 52.341465.

> Q4. What is the winpercent value for "Kit Kat"?

The winpercent value for "Kit Kat" is 76.7686.

> Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

The winpercent value for "Tootsie Roll Snack Bars" is 49.653503.

We'll install the **skimr** package and use the `skim()` function on our candy data.

```r
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12

Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

From your use of the `skim()` function use the output to answer the following:

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The columns `sugarpercent`, `pricepercent`, and `winpercent` contain continuous values. The rest of the columns are either 0s or 1s.

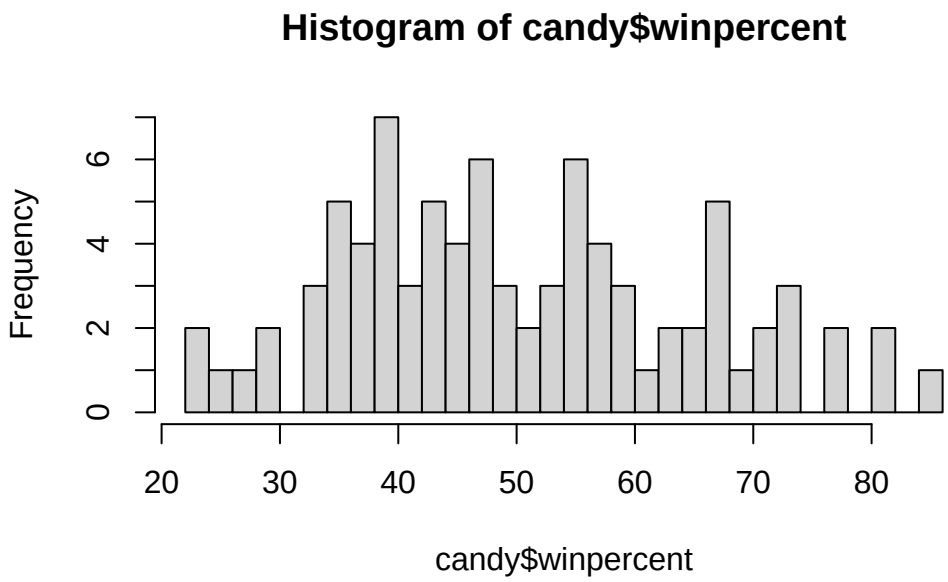
> Q7. What do you think a zero and one represent for the `candy$chocolate` column?

For the `candy$chocolate` column, a one represents a candy that contains chocolate and a zero represents a candy that does not contain chocolate.

Exploratory Analysis

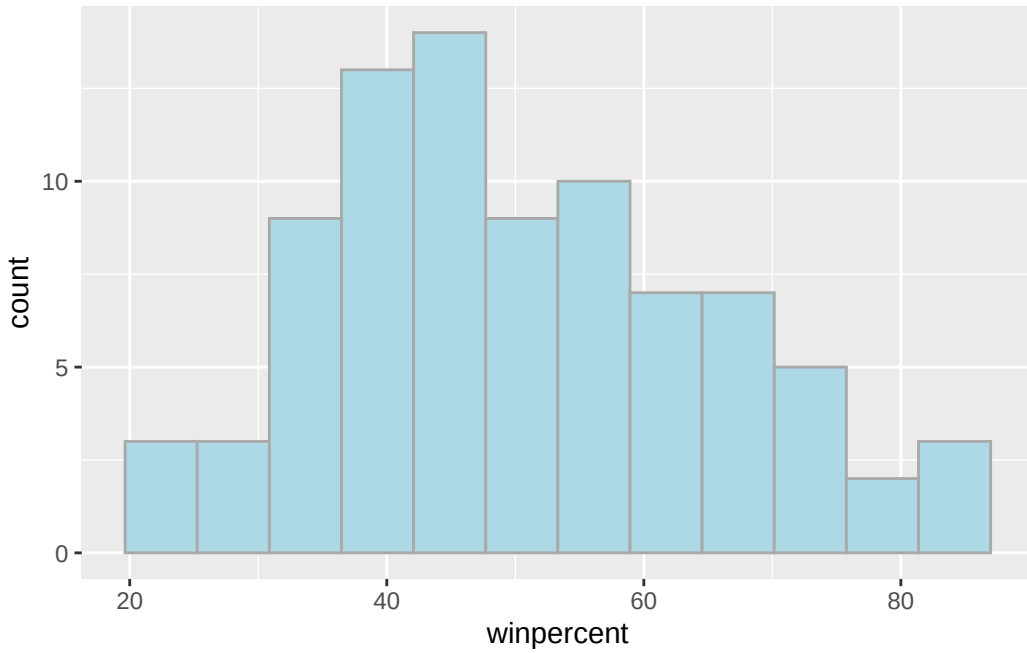
> Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
hist(candy$winpercent, breaks=30)
```




```r
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins=12, fill="lightblue", col="darkgray")
```



Q9. Is the distribution of `winpercent` values symmetrical?

The distribution of `winpercent` values is approximately symmetrical.

Q10. Is the center of the distribution above or below 50%

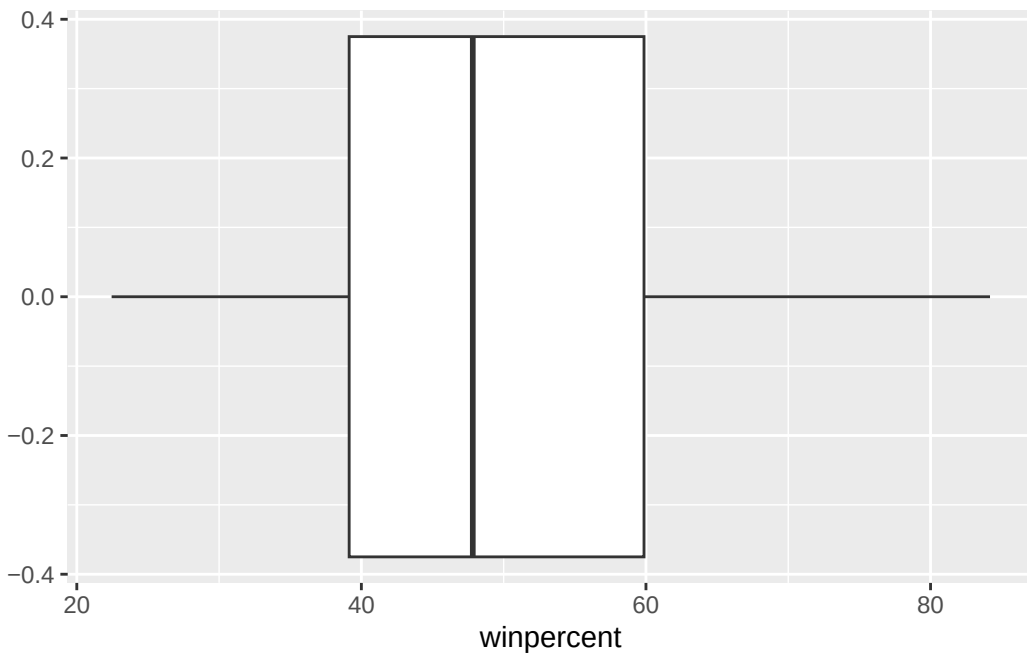
```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
summary(candy$winpercent)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  22.45   39.14   47.83   50.32   59.86   84.18 
```

```
ggplot(candy) +
  aes(winpercent) +
  geom_boxplot()
```



The mean of the distribution is above 50%, but the median of the distribution is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
choc_candy <- candy %>% filter(chocolate==1)
choc_candy_win_mean <- mean(choc_candy$winpercent)
choc_candy_win_mean
```

```
[1] 60.92153
```

```r
fruit_candy <- candy %>% filter(fruity==1)
fruit_candy_win_mean <- mean(fruit_candy$winpercent)
fruit_candy_win_mean
```

```
[1] 44.11974
```

On average, chocolate candy is higher ranked than fruit candy.

Q12. Is this difference statistically significant?

```r
t.test(choc_candy$winpercent, fruit_candy$winpercent)
```

```
    Welch Two Sample t-test

data:  choc_candy$winpercent and fruit_candy$winpercent
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Based on the the t-test's p-value of 2.871e-08, the difference in means is statistically significant.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```r
head(candy[order(candy$winpercent),], 5)
```

```
                    chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                   0      1       0              0      0
Boston Baked Beans          0      0       0              1      0
Chiclets                    0      1       0              0      0
Super Bubble                0      1       0              0      0
Jawbusters                  0      1       0              0      0
                    crispedricewafer hard bar pluribus sugarpercent pricepercent
```

```
Nik L Nip                          0    0   0        1        0.197        0.976
Boston Baked Beans                 0    0   0        1        0.313        0.511
Chiclets                           0    0   0        1        0.046        0.325
Super Bubble                       0    0   0        0        0.162        0.116
Jawbusters                         0    1   0        1        0.093        0.511
                    winpercent
Nik L Nip             22.44534
Boston Baked Beans    23.41782
Chiclets              24.52499
Super Bubble          27.30386
Jawbusters            28.12744
```

The five least liked candy types in this set are: Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

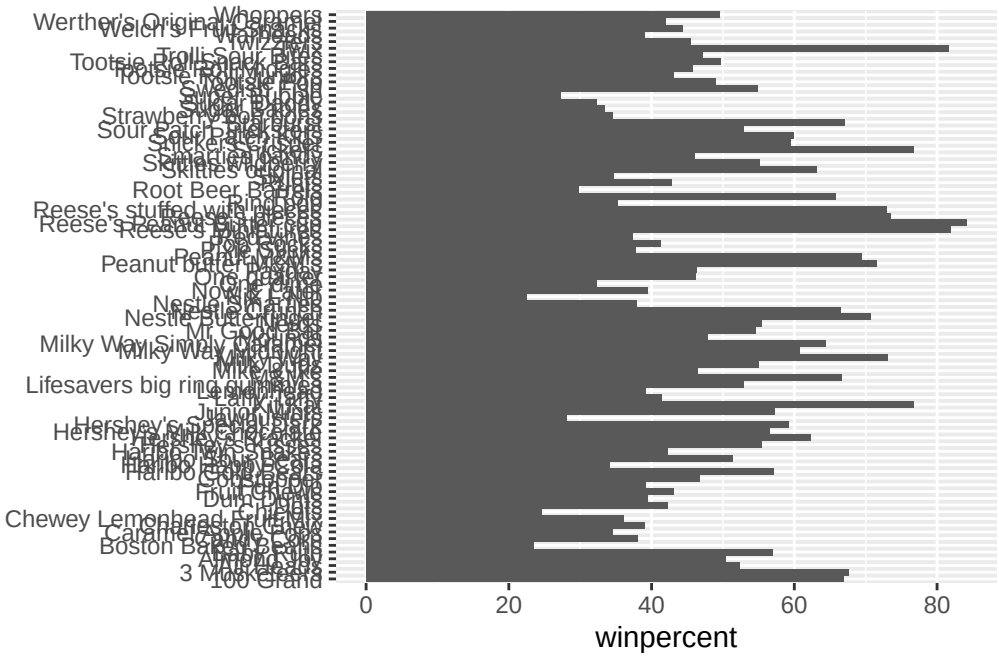
```r
head(candy[order(desc(candy$winpercent)),], 5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

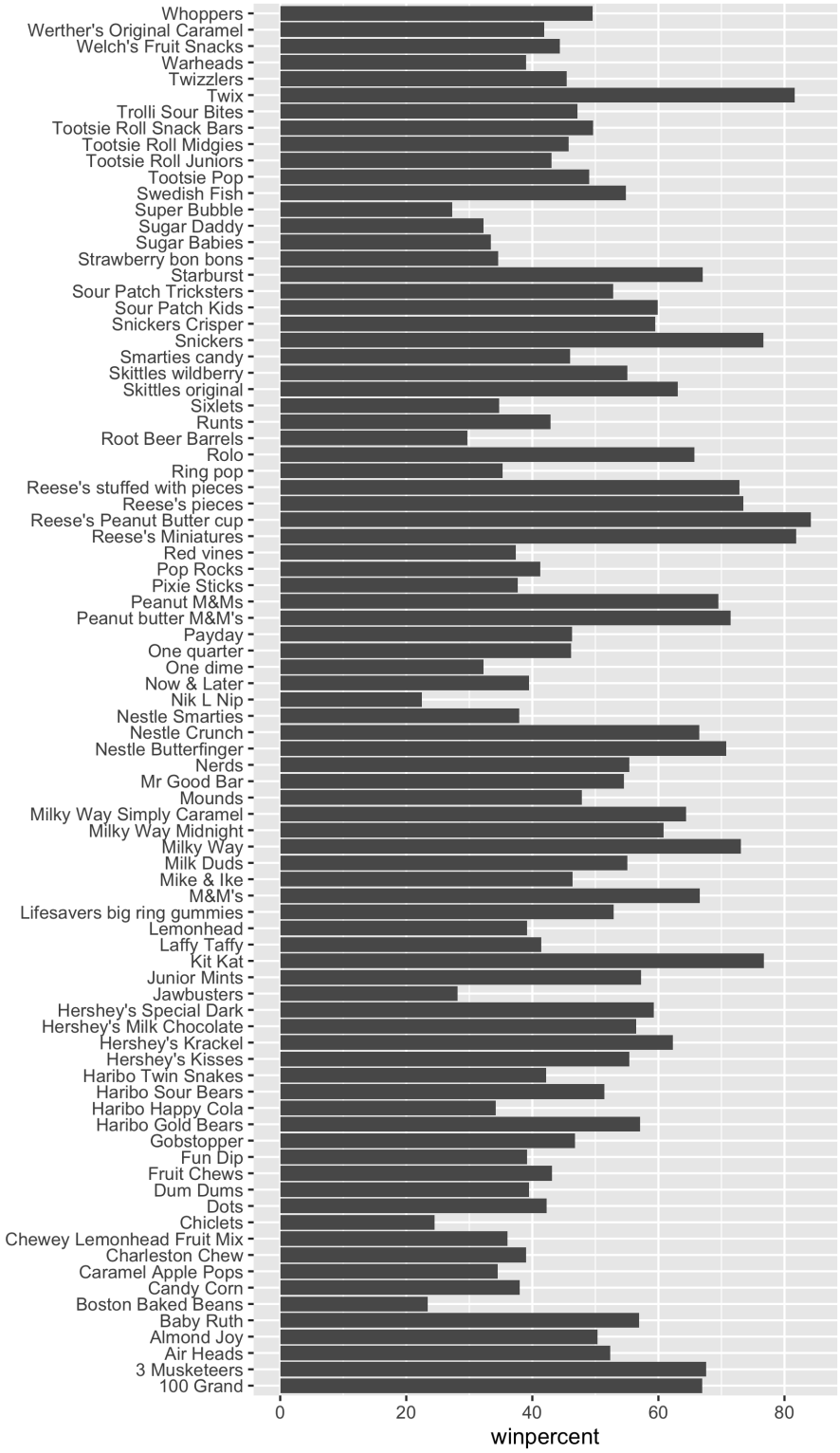
The top 5 all time favorite candy types out of this set are: Reese's Peanut Butter cups, Reese's Miniatures, Twix, Kit Kat, and Snickers.

Q15. Make a first barplot of candy ranking based on `winpercent` values.

```r
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col() +
  ylab("") # turn off Y-label that we dont need
```

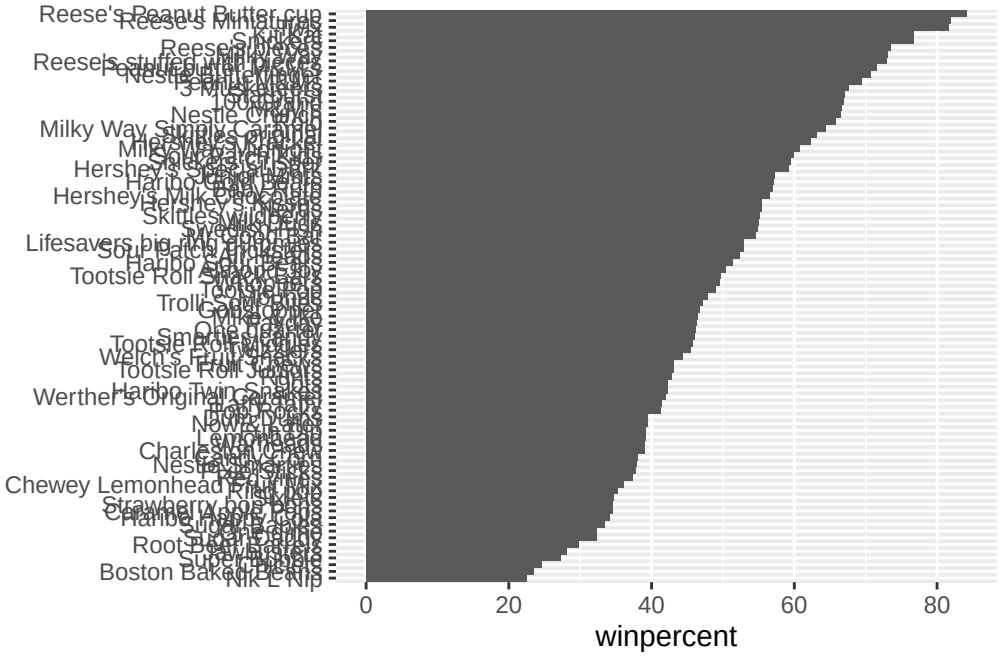



```r
ggsave("barplot1.png", height=10, width=6)
```

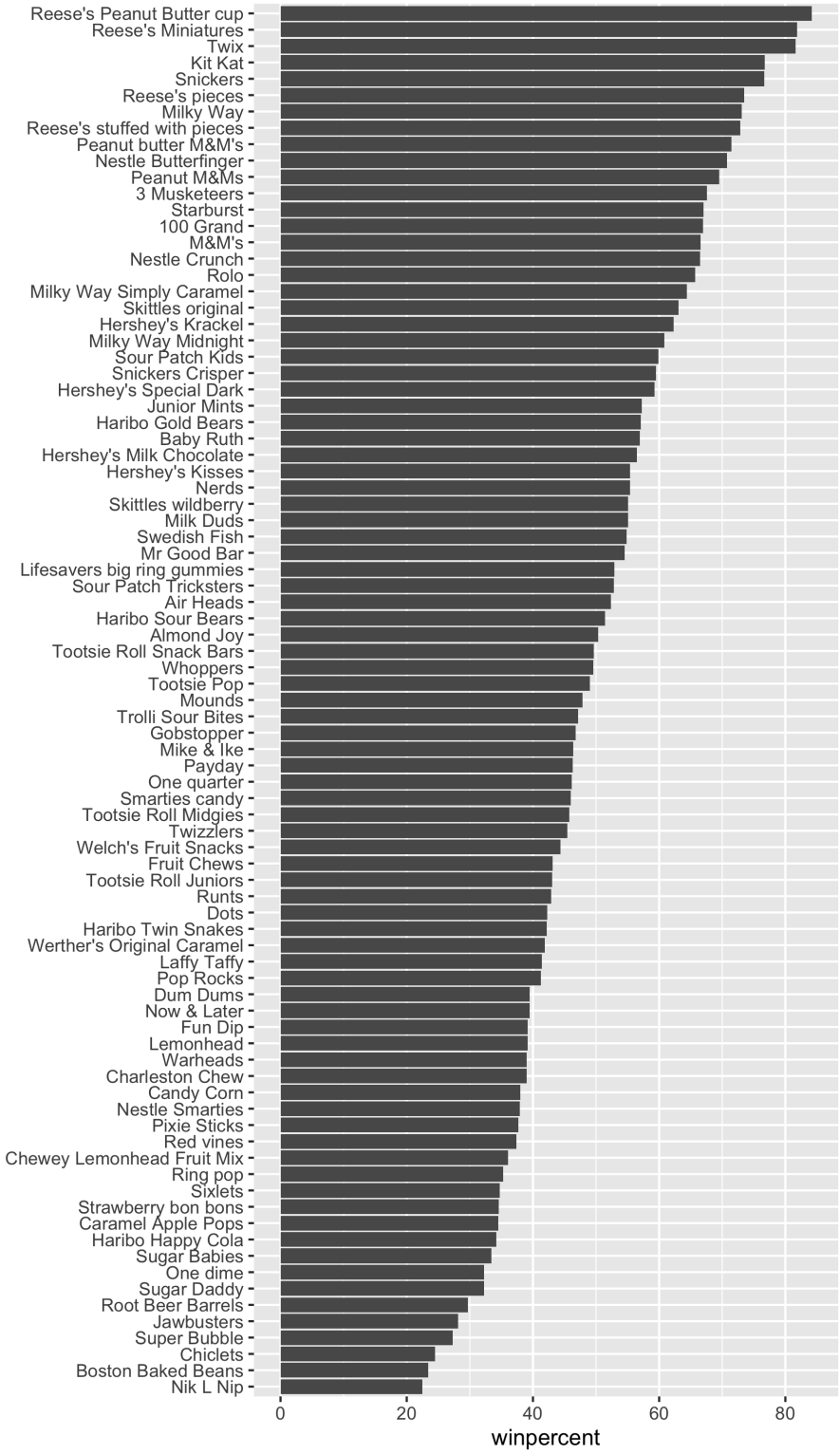

Whoppers
Werther's Original Caramel
Welch's Fruit Snacks
Warheads
Twizzlers
Twix
Trolli Sour Bites
Tootsie Roll Snack Bars
Tootsie Roll Midgies
Tootsie Roll Juniors
Tootsie Pop
Swedish Fish
Super Bubble
Sugar Daddy
Sugar Babies
Strawberry bon bons
Starburst
Sour Patch Tricksters
Sour Patch Kids
Snickers Crisper
Snickers
Smarties candy
Skittles wildberry
Skittles original
Sixlets
Runts
Root Beer Barrels
Rolo
Ring pop
Reese's stuffed with pieces
Reese's pieces
Reese's Peanut Butter cup
Reese's Miniatures
Red vines
Pop Rocks
Pixie Sticks
Peanut M&Ms
Peanut butter M&M's
Payday
One quarter
One dime
Now & Later
Nik L Nip
Nestle Smarties
Nestle Crunch
Nestle Butterfinger
Nerds
Mr Good Bar
Mounds
Milky Way Simply Caramel
Milky Way Midnight
Milky Way
Milk Duds
Mike & Ike
M&M's
Lifesavers big ring gummies
Lemonhead
Laffy Taffy
Kit Kat
Junior Mints
Jawbusters
Hershey's Special Dark
Hershey's Milk Chocolate
Hershey's Krackel
Hershey's Kisses
Haribo Twin Snakes
Haribo Sour Bears
Haribo Happy Cola
Haribo Gold Bears
Gobstopper
Fun Dip
Fruit Chews
Dum Dums
Dots
Chiclets
Chewey Lemonhead Fruit Mix
Charleston Chew
Caramel Apple Pops
Candy Corn
Boston Baked Beans
Baby Ruth
Almond Joy
Air Heads
3 Musketeers
100 Grand
0
20
40
60
80
winpercent

Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col() +
  ylab("") # turn off Y-label that we dont need
```




```
ggsave("barplot2.png", height=10, width=6)
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
0
20
40
60
80
winpercent

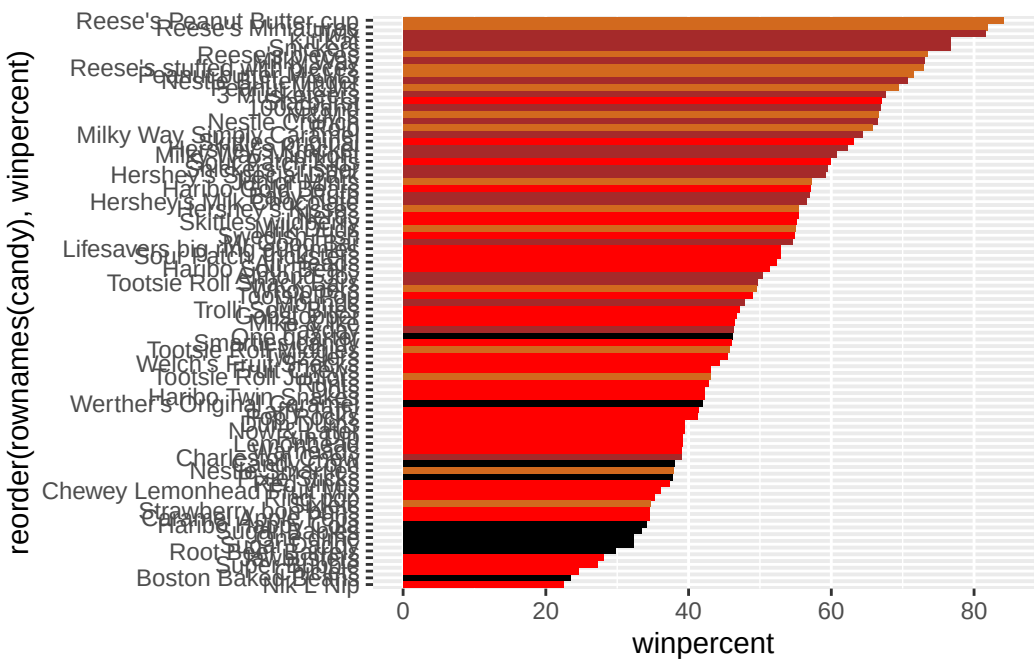
Adding color

We'll set up a color vector to symbolize candy type. We'll first make a vector of all black values. Then we overwrite the colors: chocolate for for chocolate candy, brown for candy bars, and red for fruity candy.

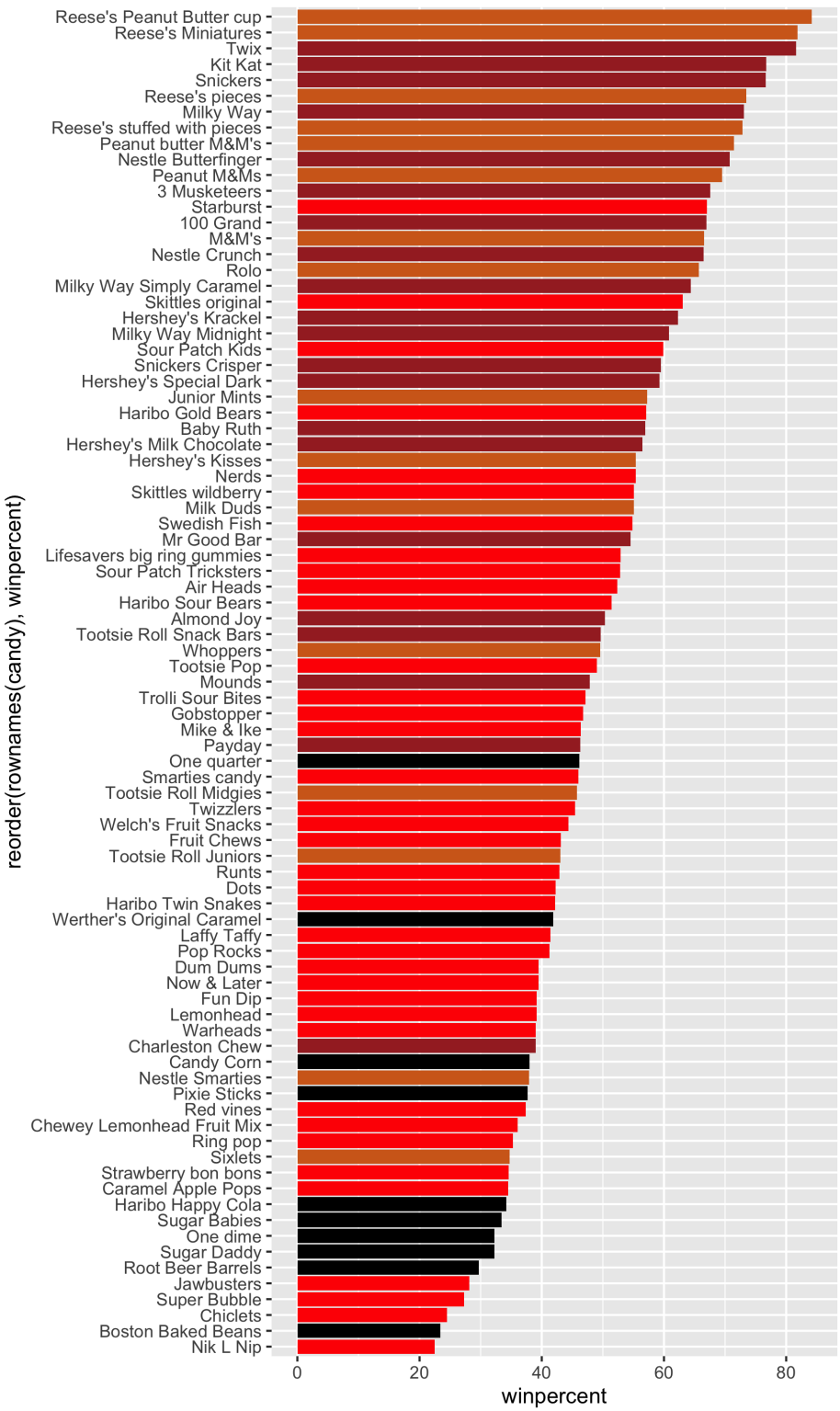
```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "red"
```

Now we'll add these colors to the barplot.

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols)
```




```
ggsave("barplot3.png", height=10, width=6)
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
reorder(rownames(candy), winpercent)
0
20
40
60
80
winpercent

Now, for the first time, using this plot we can answer questions like:

> Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is Sixlets.

> Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburst.

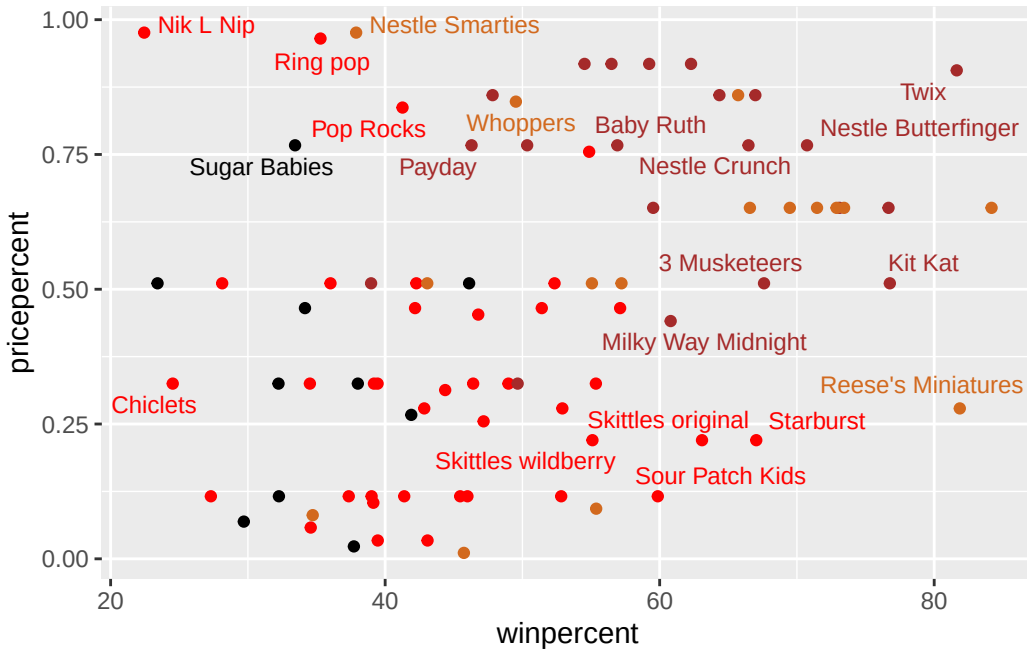
Taking a look at pricepercent

Let's analyze value for money by plotting `winpercent` vs `pricepercent`. We'll add text labels so that we can identify a given candy.

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(x=winpercent, y=pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

```
Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



Q19. Which candy type is the highest ranked in terms of `winpercent` for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures is the highest ranked in terms of `winpercent` for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing=TRUE)
head(candy[ord,c(11,12)], n=5)
```

```
                        pricepercent winpercent
Nik L Nip                      0.976   22.44534
Nestle Smarties                0.976   37.88719
Ring pop                       0.965   35.29076
Hershey's Krackel              0.918   62.28448
Hershey's Milk Chocolate       0.918   56.49050
```

The top 5 most expensive candy types in the dataset are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. Of these, the least popular is Nik L Nip.

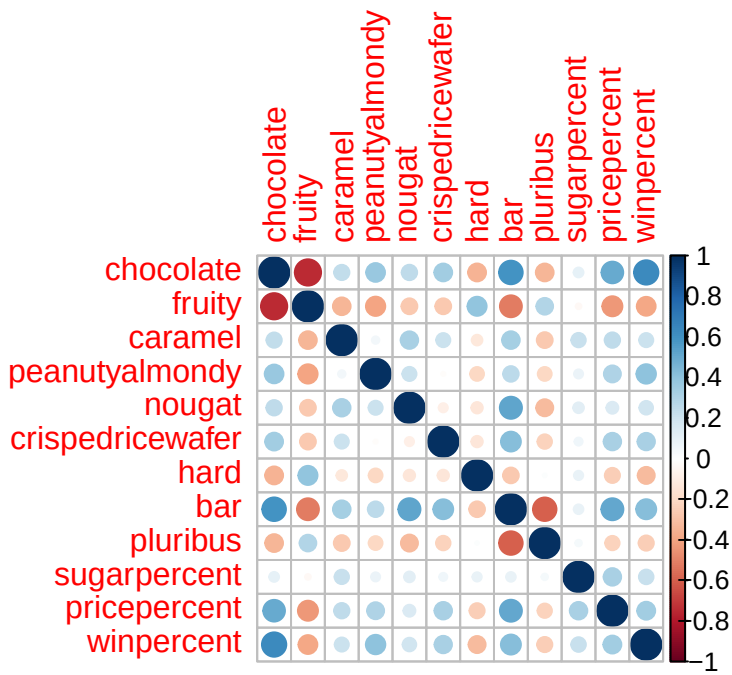
Exploring the correlation structure

Now let's see how the variables interact with one another. We'll use the **corrplot** package to plot a correlation matrix.

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



> Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The two variables that are anti-correlated are chocolate and fruity.

> Q23. Similarly, what two variables are most positively correlated?

The two variables that are most positively correlated are chocolate and bar.

Principal Component Analysis

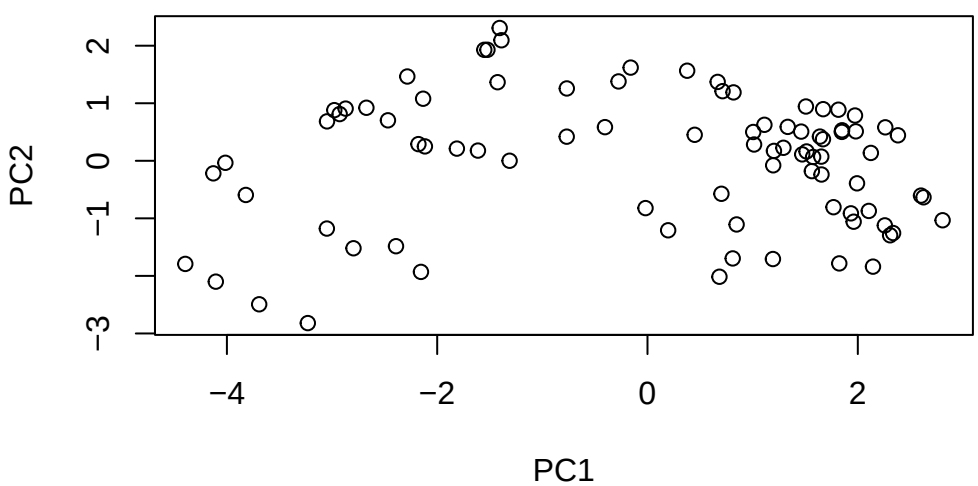
Let's apply PCA using `prcomp()` to our candy dataset. We will set `scale=TRUE`.

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

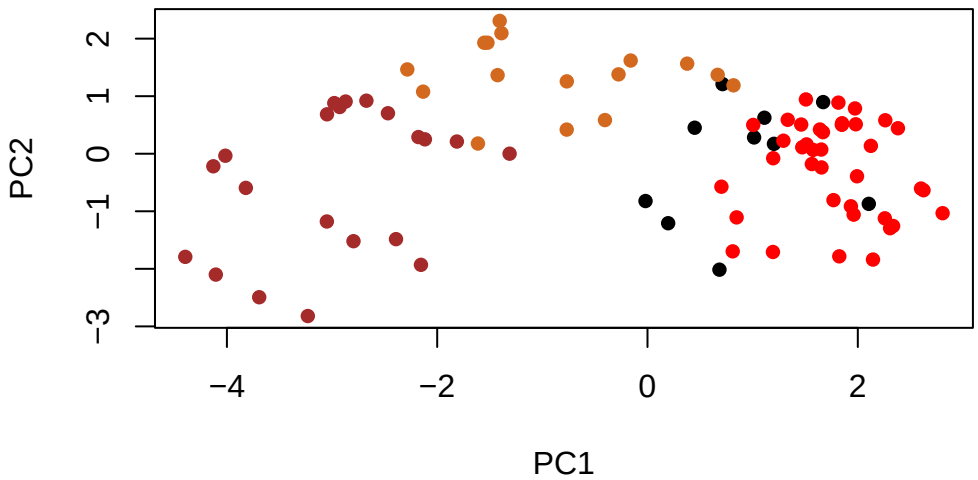
Now we'll plot our PCA score plot of PC1 vs PC2.

```
plot(pca$x[,1:2])
```



We'll change the plotting character and add color.

```r
plot(pca$x[,1:2], col=my_cols, pch=16)
```

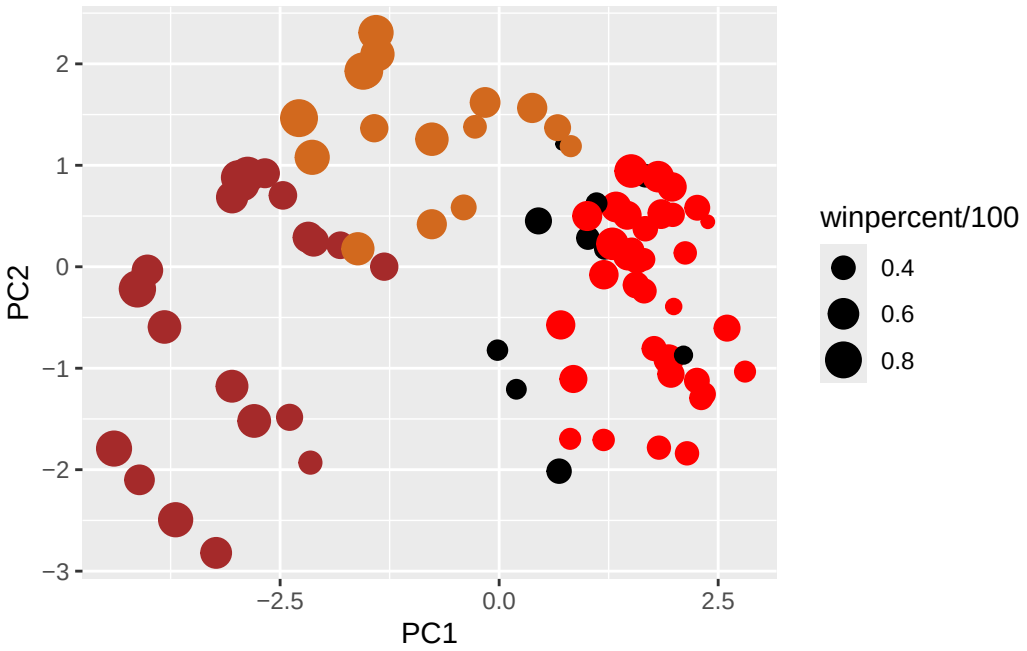


We can make a nicer plot with **ggplot2**, but we need to make a new data.frame that includes a separate column for each aesthetic to be displayed in the plot. We'll make a new data.frame containing the PCA results along with the candy data.

```r
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```r
p <- ggplot(my_data) +
        aes(x=PC1, y=PC2,
            size=winpercent/100,
            text=rownames(my_data),
            label=rownames(my_data)) +
        geom_point(col=my_cols)

p
```

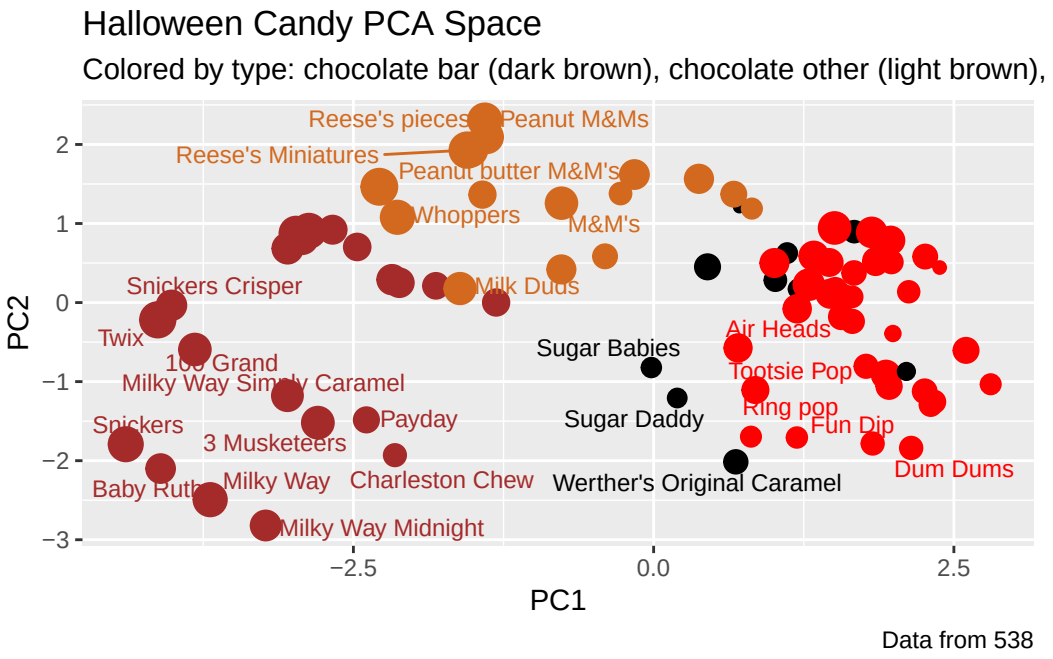


We'll use the **ggrepel** pacakge again and the function `ggrepel::geom_text_repel()` to label the plot with non-overlapping candy names. We will also add a title and subtitle.

```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7)  +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
       caption="Data from 538")
```

```
Warning: ggrepel: 59 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

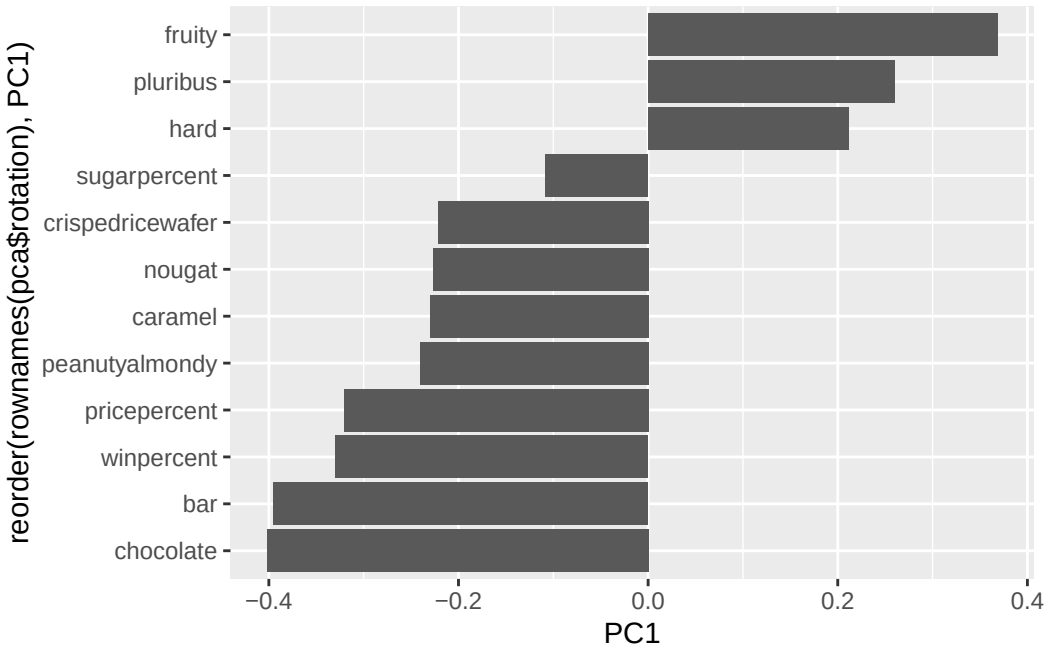


To see more candy labels we can change `max.overlaps` to allow more overlapping labels. Or, we can pass the ggplot object `p` to **plotly** to generate an interactive plot where you can hover over points to see labels.

```
#library(plotly)
#ggplotly(p)
```

Let's take a look at our PCA loadings plot.

```
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



> Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Fruity, pluribus, and hard are picked up strongly by PC1 in the positive direction. These make sense to me because these three traits are commonly seen together. This relationship was highlighted previously in the correlation matrix, with fruity and chocolate being anti-correlated.

Summary

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

The characteristics that appear to make a "winning" candy are chocolate and bar. The first barplots show chocolate candy clustered towards the top. The correlation matrix shows a positive correlation between winpercent and both chocolate and bar. Finally, the PCA analysis shows that chocolate and bar contribute most to PC1 in the negative direction. All of these analyses indicate that chocolate and bar are a winning combination.